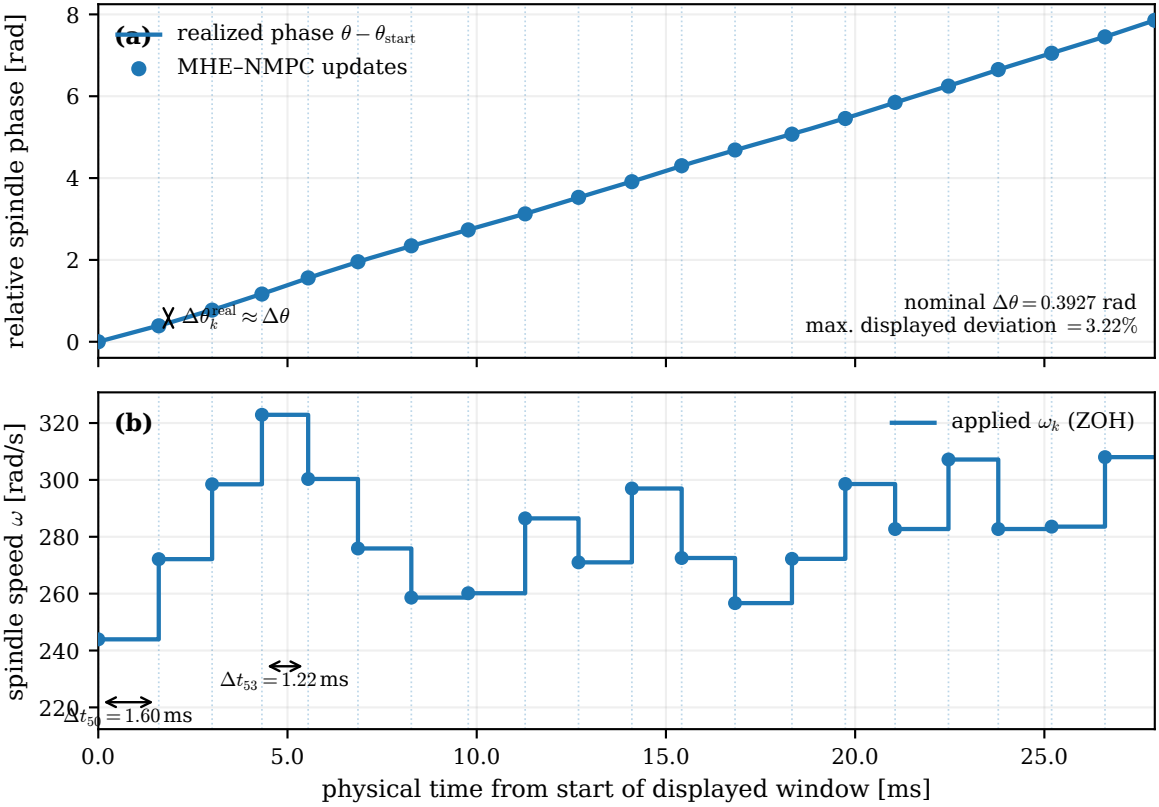
(a)
realized phase $\theta - \theta_{\text{start}}$
MHE–NMPC updates
$\Delta\theta_k^{\text{real}} \approx \Delta\theta$
nominal $\Delta\theta = 0.3927$ rad
max. displayed deviation $= 3.22\%$
relative spindle phase [rad]
0
2
4
6
8
(b)
applied ω_k (ZOH)
$\Delta t_{53} = 1.22$ ms
$\Delta t_{50} = 1.60$ ms
spindle speed ω [rad/s]
220
240
260
280
300
320
0.0
5.0
10.0
15.0
20.0
25.0
physical time from start of displayed window [ms]